Unit: 6 Unit 6: Systems of Linear Equations and Inequalities
Topic: Solving Systems Using Elimination and Substitution

Name: ______________
Date: __________

Advanced (Habanero)

Q1. Solve the system: $x + y = 4$, $x - y = 2$. What is x?
(A) 1
(B) 2
(C) 3
(D) 4
(E) 5

Q2. Solve the system: $2x + 3y = 7$, $x - 2y = -3$. What is y?
(A) -1
(B) 0
(C) 1
(D) 2
(E) 3

Q3. Solve the system: $x + 2y = 5$, $3x - 2y = 7$. What is x?
(A) 1
(B) 2
(C) 3
(D) 4
(E) 5

Q4. The system $x + y = 2$, $2x + 2y = 4$ has
(A) one solution
(B) no solution
(C) infinitely many solutions
(D) two solutions
(E) three solutions

Q5. Solve the system: $x - y = 1$, $3x + 3y = 12$. What is y?
(A) -1
(B) 0
(C) 1
(D) 2
(E) 3

Q6. The system $2x - 3y = 5$, $4x - 6y = 10$ has
(A) one solution
(B) no solution
(C) infinitely many solutions
(D) two solutions
(E) three solutions

Q7. Solve the system: $x + 2y = 7$, $2x + 4y = 12$. What is x?
(A) 1
(B) 2
(C) 3
(D) 4
(E) 5

Q8. The system $x + y = 4$, $x + y = 5$ has
(A) one solution
(B) no solution
(C) infinitely many solutions
(D) two solutions
(E) three solutions

Open Ended Questions (FRQ)

Q9. Solve the system $x - 2y = -3$, $3x + 2y = 7$ using substitution or elimination. Show all steps.

Q10. Solve the system $2x + y = 4$, $x - 3y = -5$ using substitution or elimination. Show all steps.

Chef's Tips

- Tip 1: Remind students to check their work by plugging their solutions back into the original equations.
- Tip 2: Emphasize the importance of labeling their variables andconstants clearly.
- Tip 3: Encourage students to use elimination when the coefficients of one variable are the same in both equations.